

* Bitcoin

- Cryptocurrency, Internet based digital currency, Virtual currency
- decentralized digital currency
- leverage peer to peer network over Internet.
- Miner's profit / mint bitcoin
- ~~no~~ central authority, every machine that mines bitcoins constitute part of bitcoin network
- easy to set up
- anonymous
- transparent
- Fast transfers
- No mandatory transaction fees.
- Non-repudiation

Blockchain

What is blockchain?

- a distributed ledger
- sequential block
- each block contains footprint of previous block
- a block contains transaction that can be easily verified by anybody

Blockchain enables smart contracts.

- Blockchain is a distributed database over internet
- append only, transparent, incorruptible, secure, time-stamped without inconsistent copies on different computers (consensus)

Digital Sign.

- Analogue of sign. and or.
- Should not be possible to forge
- Signing authority should not be able to claim that I did not sign.

* digital sign. is a mathematical techn. used to validate the authenticity and integrity of a message, software or digital document.

* Blind Signature

Digi Cash

1. Anit chooses serial number
2. Bank signs (blind) w/o knowing serial no.
3. While spending Anit reveals the serial no. to Mohit
4. Mohit asks for double spending
5. Bank checks in DB if the serial number exists in spent list issues double spending
6. Else add this number to in spent list
7. Mohit now issues a new note.

⑥ Cryptographic Hash Functions

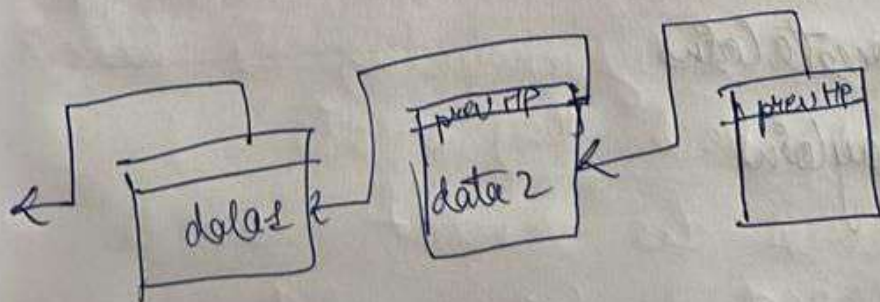
→ Desirable properties

- Deterministic
- Efficient to compute
- Difficult to find a message from its hash (pre-image resistance)
- Infeasible to find out 2 messages having same hash value (Collision resistance)
- Any small change in the message should ~~or~~ change the hash value extensively.

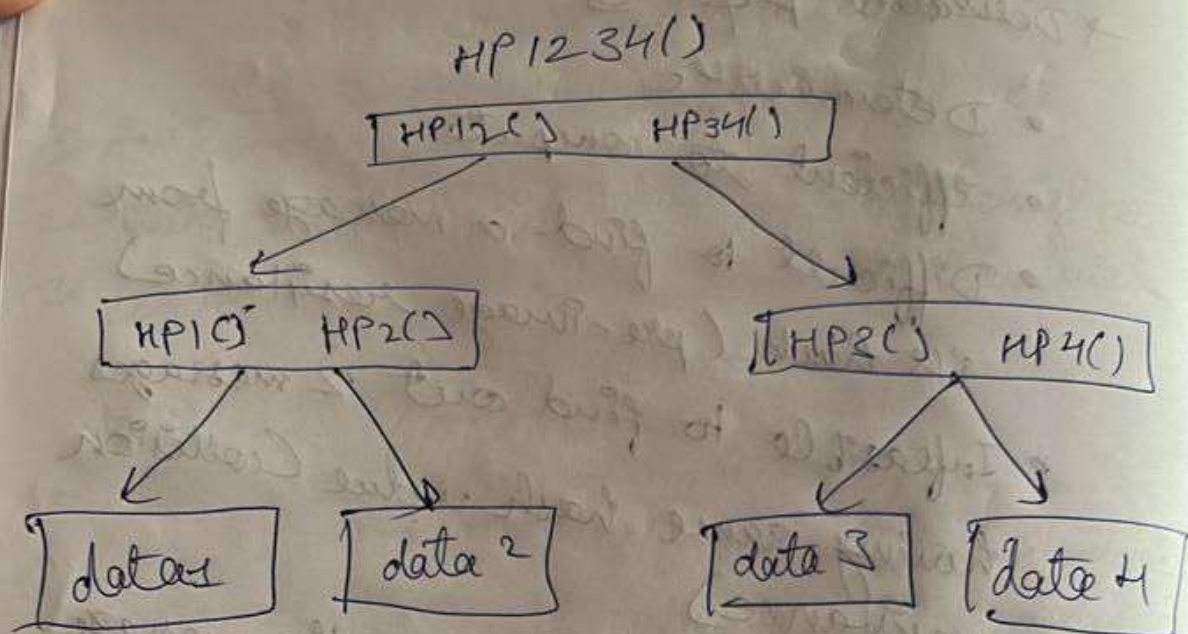
⑦ Hash Pointers

→ Pointer to the data along with its hash.

⑧ Hash Pointers to build an append only ledger



* Hash Pointer? Merkle Tree



* First Attempt to Digital Currency

- Create coins
- Signatures
- Transfer of ownership

→ we use hash pointers along with

→ signatures

→ create coins

→ pay coins



Book 2 A. Narayan's Book

(Goofy coin / Trust me coin / etc.)

problem with trust me coin

- * centralized

- * can charge large trans. fee.

- * may not add some transac.

Decentralisation

→ where to keep this ledger? cannot be single server

→ every node in the network

→ if multiple nodes can write to even copy

Distributed Consensus

- Incentives for honesty
- Indirect penalty for not following the protocol.

* Bitcoin Consensus

- Broadcast transaction to all nodes
- Each node collects transac. into a block
- A random node gets chance to write to the ledger (b.c.) It brodc. its block

Other nodes accept the block only if all transac. in it are valid

- Nodes express their accept of the block by including its hash in next block they create
- Always the longest chain is considered to be a valid

Selecting Random Node

→ each round protocol generates a certain random string and a node with identity having same bit string in its identity.

Sybil Attack:

* Challenging

to agree on a next leader who will write in async networks (Sybil attack)

* How about this?

solve XXXX puzzle and one who solve its first will append the ledger

I cannot create multiple identities and increase chances of proposing a new block.

chance of solv.

puzzle & computing power

Malicious things a random node can do

- Sybil
- denial of services
- double spending

↓
wait for multiple confirmation;
this decreases probability of
double spending

What is Purzly?

Find out nonce such that

$$H(\text{nonce} \parallel \text{prevhash} \parallel tx_1 \parallel tx_2 \parallel \dots) < \text{target}$$

256 bit.

~~target~~

$256 - k$

2

1 k bits

000

k bits

on average need 2k comp.

Current hash rate $\approx 1017 \text{ EH/s}$

* Incentive Engineering

- Miner pays the self a block reward
- Miner collects transaction fees.

Block reward

→ current 3.125 BTC

started with 50 BTC per block

protocol says every 210K transact,
reward becomes half.

$$\text{Total coins} \approx 210K \times [50 + 25 + \dots \dots]$$

$$210K \times 100$$

$\approx 210 \text{ Million coins}$

4 years to half the reward (50 mins
between
2 consecutive
block)

$$\text{target} = \frac{\text{Difficulty} \pm \text{target}}{\text{Current difficulty}}$$

next difficulty

$$= \frac{\text{prev difficulty} * 14 \text{ days}}{\text{time taken to mine last 2016 block}}$$

time taken to mine
last 2016 block

$$= \frac{(2^{16} - 1) * 208}{\text{Current difficulty}}$$

Current difficulty